SQL and DBMS Interview Notes

Relational databases, SQL queries, keys, normalization, and transactions

DBMS Fundamentals

- A database management system stores, organizes, retrieves, and manages data. A relational DBMS represents data in tables.
- A table has rows and columns. A schema describes database structures and constraints. SQL is used to define, query, and manipulate relational data.
- Interview prompts: What is a DBMS? What is the difference between DBMS and RDBMS?

Keys and Constraints

- A primary key uniquely identifies each row and cannot be NULL. A candidate key is a minimal set of attributes that uniquely identifies a row.
- A foreign key references a key in another or the same table and helps enforce referential integrity.
- NOT NULL, UNIQUE, CHECK, DEFAULT, and key constraints help maintain valid data.
- Interview prompts: Compare primary key and foreign key. What is a composite key?

SQL Query Basics

- SELECT retrieves data. WHERE filters rows. ORDER BY sorts results. GROUP BY groups rows for aggregate calculations. HAVING filters groups.
- Common aggregate functions include COUNT, SUM, AVG, MIN, and MAX.
- Example: SELECT department, COUNT(*) FROM employees GROUP BY department;
- Interview prompts: Compare WHERE and HAVING. What is GROUP BY used for?

Joins

- INNER JOIN returns matching rows. LEFT JOIN keeps all left-side rows and matching right-side rows. RIGHT JOIN keeps all right-side rows. FULL OUTER JOIN returns matched and unmatched rows from both sides where supported.
- Join conditions commonly match related key columns between tables.
- Interview prompts: Compare INNER JOIN and LEFT JOIN. Why join tables?

Normalization

- Normalization organizes relational data to reduce unnecessary duplication and modification anomalies.
- First Normal Form (1NF) requires atomic values in each row-column position and no repeating groups under the usual relational model.
- Second Normal Form (2NF) requires 1NF and no partial dependency of a non-prime attribute on part of a candidate key.
- Third Normal Form (3NF) addresses inappropriate transitive dependencies of non-key attributes on keys.
- Interview prompts: What is normalization? Explain 1NF, 2NF, and 3NF.

Transactions and SQL Practice

- A transaction is a logical unit of database work. ACID stands for Atomicity, Consistency, Isolation, and Durability.
- COMMIT makes a transaction's changes permanent; ROLLBACK undoes uncommitted changes where supported.
- Practice queries: find the second-highest salary, count employees by department, identify duplicates, and join employee and department tables.
- Test queries against sample data and explain how NULL values affect comparisons and aggregate results.